

Muhammad Kaif Laghari

Bachelor of Science in Software Engineering
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Summary

Final-year Software Engineering student bridging practical DevOps experience with Software Quality Assurance (SQA). Skilled in automating application delivery through CI/CD pipelines and designing comprehensive manual testing suites for robust system validation. Brings hands-on expertise in AWS architecture and container orchestration (Docker, Kubernetes) to ensure highly reliable, secure, and quality-driven deployments.

Education

Bachelor of Science in Software Engineering

Shaheed Zulfiqar Ali Bhutto Institute of Science and Technology

August 2022 - May 2026

CGPA: 3.46

Technical Skills

Programming Language: Python & Flask

Cloud Platform: Amazon Web Services (AWS): EC2, S3, VPC, IAM, ALB, CodeDeploy, CloudWatch

Containerization & Orchestration: Docker, Kubernetes, GitHub Actions

Version Control: Git, GitHub

Professional Experience

DevOps Intern

CloudLem - Internship ([Letter of Recommendation](#))

Nov 2025 - Dec 2025

Gilgit Remote, Pakistan

- **Infrastructure & Security:** Designed deployment architectures and enforced security best practices using GitHub Secrets to protect production environments.
- **Server Configuration & High Availability:** Maintained reliable environments for containerized apps, ensuring high availability and efficient image distribution via DockerHub.
- **Monitoring & Incident Prevention:** Automated pipelines using GitHub Actions, and proactively monitored and resolved deployment issues to prevent incidents and maintain system stability.

Publications & Research

Native Mixed Reality Compositing on Meta Quest 3 - ([ArXiv](#))

A Quantitative Feasibility Study of ARM-Based SoCs and Thermal Headroom

Sep 2025

- Evaluated native MR compositing feasibility on the Meta Quest 3 to untether content creation from external PCs. Benchmarked the thermal and processing constraints of the Snapdragon XR2 Gen 2 against competing ARM SoCs, establishing crucial performance baselines for future standalone, high-fidelity MR systems.

Academic & Technical Projects

Automated CICD with AWS CodeDeploy & GitHubActions

Personal Project ([Project Link](#))

- Built an end-to-end CI/CD pipeline using GitHub Actions and AWS CodeDeploy to automate the build, test, and deployment of a Dockerized Flask application.
- Automated Docker image creation and versioning with GitHub Actions workflows, integrating secure deployment pipelines using AWS S3 and EC2 instances.
- Integrated AWS S3 and CodeDeploy for storing build artifacts and executing zero-downtime deployments to EC2 instances.

Software Quality Assurance: RBAC Manual Testing Suite

Personal Project ([Project Link](#))

- Designed a manual testing suite to validate Role-Based Access Control (RBAC) and security boundaries for an enterprise HR system.

- Authored a comprehensive test execution log with 11 structured cases, applying boundary value analysis and equivalence partitioning.
- Executed negative security testing and documented simulated privilege escalation vulnerabilities in a formal defect report.

Optimizing Traffic Distribution with AWS ALB

Personal Project ([Project Link](#))

- Implemented AWS ALB to distribute incoming traffic across multiple EC2 instances, improving application performance and reducing latency.
- Set up ALB to distribute traffic to multiple EC2 instances based on routing rules.
- Improved application availability by distributing incoming traffic to healthy EC2 instances.
- Applied auto-scaling policies to dynamically adjust the number of EC2 instances based on load.

Certifications

Introduction to Containers w/ Docker, Kubernetes & OpenShift - ([IBM/Coursera](#))

2025

- Learned to build and deploy applications using Docker, Kubernetes, and OpenShift.
- Gained expertise in leveraging YAML for container orchestration and managing cloud-native applications.

AWS Cloud Technical Essentials - ([AWS/Coursera](#))

2025

- Gained foundational knowledge of AWS core services including EC2, S3, RDS, IAM, Lambda & CloudWatch.
- Learned how to architect basic cloud infrastructure using AWS best practices for scalability, security, and cost-efficiency.
- Practiced navigating the AWS Management Console and using the AWS CLI for resource provisioning.